

1. Introduction

This project focuses on a binary classification problem using the Spambase dataset^[1], with the aim of comparing two widely used methods in neural computing: Support Vector Machines (SVM) and Multilayer Perceptrons (MLP). The dataset consists of 57 features extracted from email content, where the task is to predict whether each instance is spam or non-spam^[5]. The choice to compare these algorithms is justified by their different approaches to learning. SVM shows excellent results when dealing with high dimensional datasets and is capable of constructing non-linear decision boundaries by means of the kernel functions^[2]. At the same time, MLP models are examples of neural networks that are able to learn highly non-linear relationships by using a multi-layer architecture, although proper and careful tuning and training is required^[3]. The main objective of this study is not only to compare the predictive performance of these models, but also to analyse their behaviour during training, their stability, and their ability to generalise to unseen data. To ensure a fair comparison, a structured evaluation strategy is followed, including separate training, validation, and test sets, along with cross-validation and repeated experiments where appropriate.

2. Methods

2.1 Support Vector Machine (SVM):

Support Vector Machine (SVM) is one of the supervised learning models that aim to find an optimal decision boundary that separates classes by maximising the margin between them^[2]. For instances where the data does not show linear separability, the use of kernel functions enables the projection of the data into higher dimensions with a view to obtaining a non-linear decision boundary^[2]. In the current experiment, an SVM with a Radial Basis Function (RBF) kernel was used, as it is well-suited for handling non-linear relationships in high-dimensional data. The model's performance depends heavily on the choice of hyperparameters, particularly the regularisation parameter (C) and the kernel parameter (gamma), which control the trade-off between margin maximisation and classification error, as well as the flexibility of the decision boundary. To identify an appropriate configuration, a set of values for C (0.1, 1, and 10) and gamma (scale, 0.01, and 0.1) were explored using 5-fold stratified cross-validation. This approach ensured that class proportions were preserved across folds and allowed for a more reliable estimate of model performance. The final model was selected based on mean validation accuracy and consistency across folds.

2.1.1 Pros and Cons

One advantage of SVMs in this context is their deterministic nature, meaning that repeated training with the same data and parameters produces identical results. This makes the model stable and easier to evaluate. In addition, SVMs are effective in high-dimensional spaces and can model non-linear relationships through the use of kernel functions.

However, SVMs can be sensitive to the choice of hyperparameters, particularly C and gamma, which can significantly affect performance. They also do not provide insights into training dynamics, as the optimisation process is not iterative in the same way as neural networks. This can make it harder to analyse learning behaviour compared to models such as MLPs.

2.2 Multilayer Perceptron (MLP):

A Multilayer Perceptron (MLP) is a feedforward neural network that learns a mapping between inputs and outputs through multiple layers of interconnected neurons, with parameters optimised using backpropagation^[3]. In each layer, a linear operation is followed by a non-linear activation, enabling the neural network to fit complex patterns from the input data. In this study, the MLP was designed for a binary classification task with 57 input features. Different architectures were explored, including models with two and three hidden layers. Based on validation performance, a three-layer architecture was selected. This configuration had an input layer with 57 neurons, and hidden layers with 64, 32, and 16 neurons, respectively, and a single neuron for output. ReLU activation functions were used in the hidden layers, while a sigmoid activation function was applied at the output layer. The model was trained using binary cross-entropy loss. In addition to architectural choices, different optimisation strategies were considered, including Adam, Adam with weight decay, SGD with momentum, and SGD with momentum and weight decay. Comparing their performance during the model selection process, SGD with momentum was chosen as the final optimizer, with a learning rate of 0.01.

2.2.1 Pros and Cons

One of the main advantages of MLPs is their ability to model complex non-linear relationships through multiple layers and activation functions. This flexibility allows them to adapt well to datasets where patterns are not easily captured by simpler models. Increasing the number of hidden layers can improve the model's capacity to learn more detailed representations of the data.

However, MLPs require careful selection of architecture and optimisation parameters, as performance can be sensitive to these choices. The training process is also influenced by randomness, meaning that results can vary across runs. In addition, compared to simpler models such as SVMs, MLPs are generally more computationally demanding and less interpretable.

2.3 Hypothesis

Based on the characteristics of the dataset and the properties of the models, it is expected that both SVM and MLP will perform well on this classification task, but with different strengths. The SVM, particularly with an RBF kernel, is expected to provide stable and consistent performance due to its ability to handle high-dimensional data and its deterministic training process. In contrast, the MLP is expected to learn more complex non-linear relationships through its multi-layer structure, which may lead to improved predictive performance. However, this is also expected to introduce greater sensitivity to parameter choices and variability across training runs.

3. Methodology

3.1 Data Splitting

The dataset was divided into three subsets: training, validation, and test sets. The training set was used for model fitting. For both the SVM and MLP, hyperparameter tuning and model selection were carried out using 5-fold stratified cross-validation on the training set, where each fold was split into a training subset and a validation subset for performance estimation. In addition, for the MLP, a separate validation set was used during model development to support the selection of optimisation strategies, including choices such as momentum and weight decay. Models were trained on the training set and evaluated on this validation set to compare different configurations. The test set was held out and used only once for the final evaluation of the selected models, ensuring an unbiased estimate of performance. To support reproducibility, the test dataset was saved separately and reused during evaluation without any modification.

3.2 Data Preprocessing

Feature scaling was applied using a StandardScaler. The scaler was fitted only on the training data and then applied to the test set, ensuring that no information from the test data influenced the training process. Dataset was split into training and test sets using an 80/20 ratio, with stratified sampling to preserve class distribution. A fixed random state (42) was used to ensure reproducibility.

3.3 Hyperparameter Tuning and Model Selection

For both the SVM and MLP models, hyperparameter tuning was initially carried out using 5-fold stratified cross-validation on the training set. For the SVM, different values of the regularization parameter (C) and kernel parameter (gamma) were evaluated, and the final model was selected based on

mean validation accuracy and consistency across folds.

For the MLP, in addition to cross-validation, a separate validation set was used to further assess model performance. Multiple configurations were explored, including different network architectures and optimization strategies. Since MLP training is affected by randomness, each configuration was trained multiple times^[4]. Performance was assessed using mean validation accuracy and standard deviation, allowing both performance and stability to be considered during model selection.

3.4 Evaluation Metrics

The final models were evaluated on the test set using standard classification metrics, including accuracy, precision, recall, and F1-score. A confusion matrix was also used to provide a more detailed view of model performance, particularly in terms of misclassification patterns. These evaluation measures were chosen to ensure a balanced assessment of performance, considering both overall accuracy and class-specific behaviour.

4. Experimental Results

4.1 Overall Performance

The performance of the final SVM and MLP models was evaluated on the test dataset using standard classification metrics. The comparison (Table 1) focuses on class-specific performance along with overall evaluation measures. As the task focuses on identifying spam emails, particular attention is given to the performance metrics of the spam class. Here, both models achieve strong performance on the test set, with the MLP showing slightly higher values across most evaluation metrics.

Metric	Accuracy	Precision	Recall	F1-score	Macro Avg (Precision)	Macro Avg (Recall)	Macro Avg (F1)
MLP	0.933	0.92	0.91	0.91	0.93	0.93	0.93
SVM	0.921	0.91	0.88	0.90	0.92	0.91	0.92

Table 1: Performance Table of MLP and SVM

4.2 Confusion Matrix Analysis

	MLP	SVM
True Negatives	530	528
False Positives	28	30
False Negatives	34	43
True Positives	329	320

Table 2: Confusion Matrix Comparison

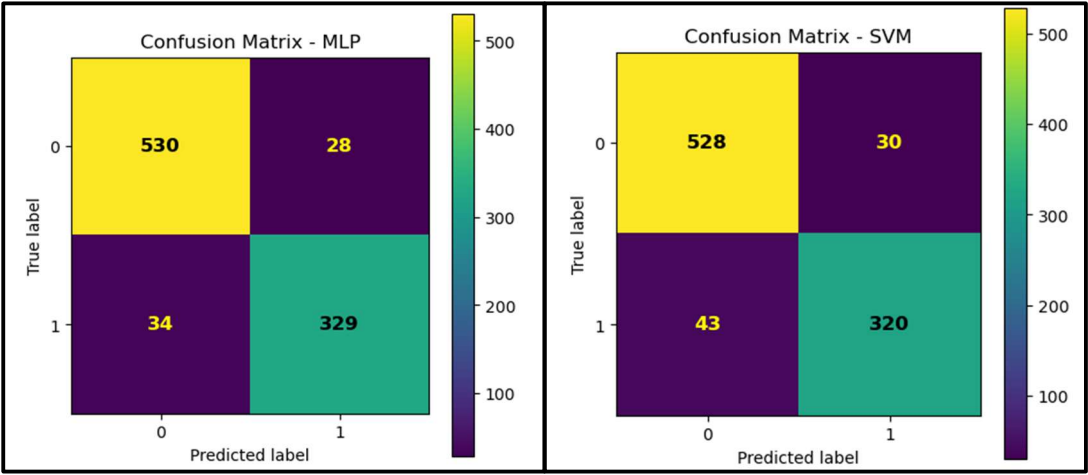


Fig 1: Confusion Matrix of MLP

Fig 2: Confusion Matrix of SVM

To further analyse model performance, confusion matrices were generated for both models to provide a detailed breakdown of classification performance. From Fig 1 and 2, the confusion matrices show that both models correctly classify the majority of instances, with relatively low numbers of false positives and false negatives.

4.3 MLP Training Behaviour

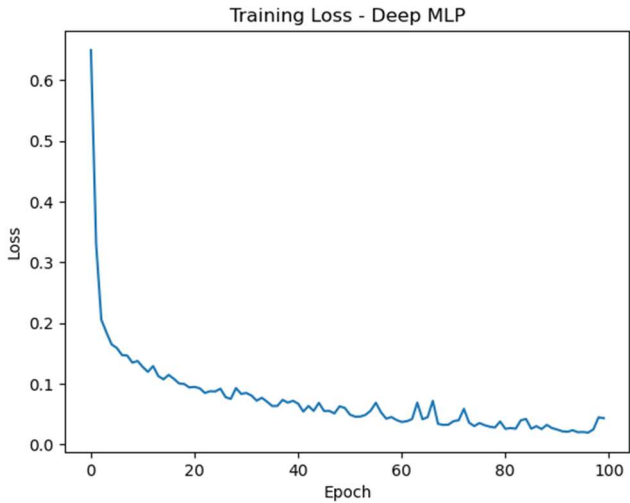


Fig 3: Training Loss Curve (MLP)

The training behaviour of the MLP model was analysed using the loss curve across epochs. In Fig 3, The loss curve shows a consistent decrease during training, indicating that the model is learning effectively and converging over time.

4.4 Model Comparison

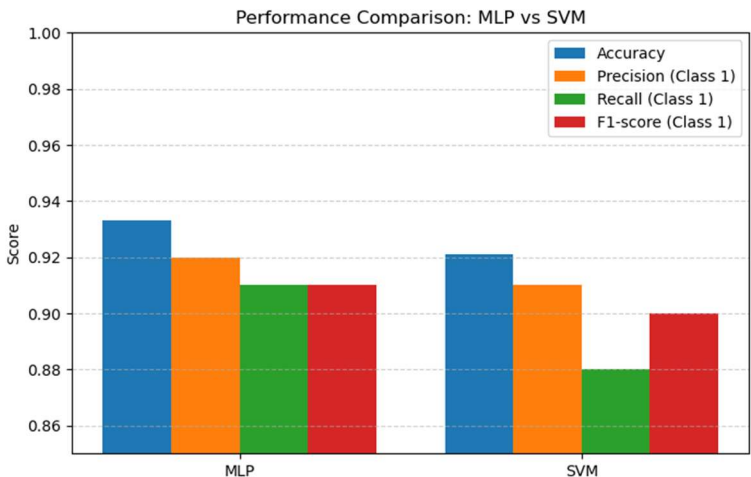


Fig 4: Performance Comparison Chart

A direct comparison of the two models in Fig 4 shows that the MLP achieves slightly higher overall performance compared to the SVM. The difference

is particularly noticeable in recall for the positive class, where the MLP correctly identifies more spam instances.

5. Critical Evaluation

5.1 Overall Performance

As shown from the experimental findings, both MLP and SVM have proven to be quite effective when applied to the current data set. In terms of overall accuracy, MLP was found to yield higher performance than SVM. The last classifier attained a score of 93.27%, compared to the 92.07% attained by the latter. While it is evident that MLP is superior to SVM in terms of predictive power, the difference in performance is quite minimal. It becomes clear when one observes the metrics of class-wise performance that MLP has marginal improvements in precision and recall scores for the positive class. For instance, while recall scores for SVM and MLP were 0.88 and 0.91, respectively, which means that the former did better in correctly recognizing positive cases. Nonetheless, the difference in performance is quite minimal and thus does not signify significant superiority on behalf of MLP. On the whole, while MLP proved to perform the best, SVM cannot be considered ineffective. The similarity in performance suggests that the dataset may not require highly complex models to achieve strong predictive results, and that simpler methods can perform comparably well.

5.2 Class-wise Behaviour & Error Analysis

Both models show an almost equally balanced distribution regarding precision and recall in both classes. Nevertheless, when considering the errors made by each classifier, one can detect a difference in their distribution. From the confusion matrices, the MLP produces fewer false negatives compared to the SVM (34 vs 43). This indicates that the MLP is more effective at correctly identifying positive instances, which is consistent with its higher recall for class 1 (0.91 compared to 0.88 for the SVM). In practical terms, this means that the MLP misses fewer positive cases, which may be important depending on the application context. However, this improvement is accompanied by a slightly higher number of false positives for the MLP (the MLP produces slightly more false positives than the SVM (30 vs 28), although the difference is minimal). This reflects a common trade-off between precision and recall, where improving the detection of positive instances can lead to a small increase in incorrect positive predictions. Thus, the differences in error distribution between the two models are not very significant, but they provide useful insight into their behaviour. The MLP demonstrates a slight advantage in detecting positive instances, while both models maintain similar overall error rates.

5.3 Model Complexity vs Performance Trade-off

MLP, with three hidden layers and multiple tuneable parameters, represents a more complex model capable of learning non-linear feature interactions. In contrast, the SVM with an RBF kernel provides a comparatively simpler approach, relying on a fixed optimisation framework with fewer hyperparameters. Despite this difference in complexity, the performance gain achieved by the MLP is relatively small. While the MLP achieves the highest accuracy, the improvement over the SVM is marginal (approximately 1%). In other words, although being much more complex, the MLP brings very little benefit for the current problem in terms of better predictions. Furthermore, the SVM demonstrates strong performance without requiring extensive tuning or iterative training, indicating that it is well-suited to the underlying structure of the data.

5.4 Limitations of the Analysis

Firstly, the model selection for the MLP involved multiple stages, including cross-validation followed by further optimisation using a separate validation split. While this approach improved performance, it also introduces a risk of overfitting to the validation data. The high variance between extremely good validation accuracy (~0.9938) and poor test accuracy (~0.9327) supports this statement. Secondly, the analysis revealed that the results are highly sensitive to preprocessing consistency. During the implementation, inconsistencies in feature scaling led to significant degradation in model performance. As a result, the reliability of the results depends on maintaining a strictly consistent preprocessing pipeline, and any deviation can lead to misleading conclusions. Finally, the comparison between models is based on a single dataset and a specific train-validation-test split. While both models perform well under these conditions, the conclusions drawn may be dependent on this particular split and may not fully generalise to other datasets or distributions.

5.5 Final Evaluation and Comparison

In general, both classification algorithms show good results with the MLP being the one to outperform others, which proves high prediction accuracy in classifying data. However, the improvement provided by the MLP over the SVM is relatively small, indicating that both models are effective for this dataset. The MLP is a bit more successful in recognising relevant observations, as reflected in its higher recall and lower number of false negatives. This suggests that it is marginally more effective in detecting important cases. However, this benefit must be considered alongside the increased complexity of the model and the additional effort required for tuning and optimisation. In contrast, the SVM achieves comparable performance with a simpler training process and more stable behaviour across different evaluation stages. Its performance remains consistent between cross-validation and test results, indicating reliable generalisation without extensive tuning.

Based on all the above-mentioned, it should be concluded that the MLP is the best classifier in relation to prediction accuracy, but the SVM should also be noted due to its competitive performance and simplicity and stability. Therefore, the choice between the two models depends on the relative importance of maximising performance versus maintaining a simpler and more robust modelling approach.

6. Lessons Learned and Future Work

This study shows that increasing model complexity does not necessarily result in a significant improvement in performance. Although the MLP achieved the highest accuracy, the improvement over the SVM was small, indicating that both models are well-suited to the dataset. The experiments also highlight that the MLP is sensitive to training configuration. The choice of optimizer had a clear impact on performance, as the model selected through cross-validation (Adam) was later outperformed by SGD with momentum during further validation. This demonstrates that model selection based on a single stage of evaluation may not always reflect final performance. Another important observation is the role of cross-validation in the modelling process. In this study, cross-validation was used as a model selection step prior to final training, rather than as a post-training evaluation method. This highlights that cross-validation is primarily a tool for selecting model configurations, and that final model performance should still be validated on a separate test set. Finally, the difference between validation and test accuracy for the MLP indicates that validation performance can be optimistic, particularly after multiple rounds of tuning. This emphasises the importance of evaluating models on a separate test set. For future work, a more systematic approach to hyperparameter tuning could be applied to ensure a more complete search for model configurations. In addition, techniques such as early stopping or alternative regularisation methods could be explored to improve the generalisation of the MLP. Evaluating the models across multiple data splits would also help to confirm the stability of the results. Finally, exploring alternative models, such as ensemble methods, could offer further improvements and provide a broader comparison of approaches.

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